



KairosAI

Real-Time Network Threat Detection & Explainable Security

Detects suspicious network activity and converts complex security events into clear, actionable intelligence.

- ✓ Real-Time NIDS (Scapy/PyShark)
- ✓ Feature & Rule Detection
- ✓ Plain-English XAI (Llama 3)

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3

Focused Attacks (Port Scan, SYN Flood, ARP Spoof)



3-Stage

Core Pipeline (Detect → Explain → Visualize)



Local AI

Llama 3 / Ollama Explanation Layer

02 The Local Network Security Challenge

Vulnerabilities in Educational and Small Organization Networks

🛡️ Core Pain Points for Campus & Local Networks

1 Alert Overload Without Context

Legacy IDS solutions spit out thousands of raw, cryptic network log entries daily that are difficult for IT staff to filter and parse.

2 Difficulty Interpreting Network Activity

Analyzing raw TCP/IP packet captures or hex streams requires deep forensic training, making it hard to quickly pinpoint root causes.

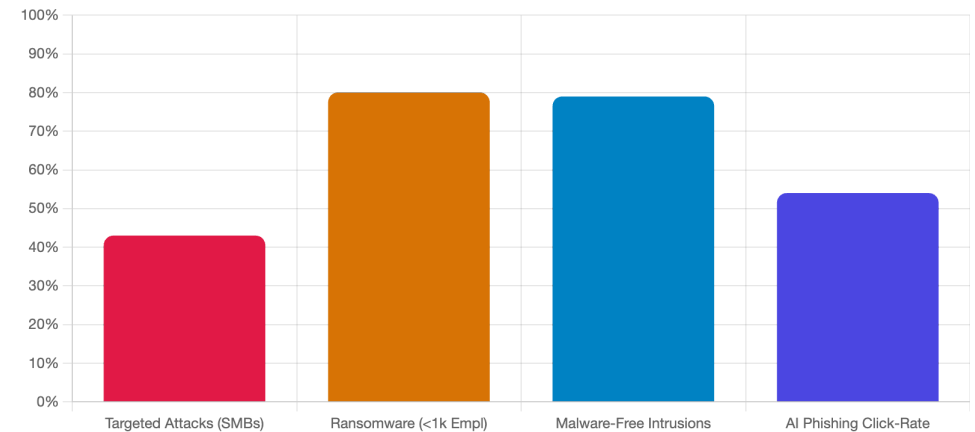
3 Limited Cybersecurity Expertise

Small school and organization IT departments generally have limited security staff, making continuous manual threat monitoring unfeasible.

4 Slow Identification of Suspicious Behavior

Without real-time behavioral telemetry, scanning or spoofing attacks can go undetected for weeks, risking data integrity.

📊 Campus Network Threat Frequency (Probes)



📌 Campus subnets encounter high rates of automated scans and MAC/IP manipulation attempts due to diverse and shared network endpoints.

🎓 **Campus Network Context:** Campus networks contain student systems, labs, administrative systems and shared devices, making fast identification of abnormal activity important.

03 Round 1 Feedback → Round 2 Focus

Narrowing Threat Scope for Reliable and Verifiable Detection

Round 1 Scope

Broad Network Threat Detection

- 1 Ambitious Multi-Vector Coverage**
Attempted to capture all classes of malware, intrusions, and anomalous exploits simultaneously.
- 2 Unfocused Feature Engineering**
Broad features led to higher false positives and limited model specificity on individual attack patterns.
- 3 Complex Live Verification**
Difficult to simulate and prove end-to-end detection rates across generic network traffic streams.

Round 2 Refinement

Focused Detection of 3 Attacks

- 1 Port Scanning**
Detect reconnaissance scan patterns across multiple ports using specialized network features.
- 2 SYN Flood DDoS**
Capture volumetric TCP SYN connection rates and analyze incomplete handshake behaviors.
- 3 ARP Spoofing**
Inspect conflicts and changes in local IP-MAC address mappings in real time.

 **Rationale for Narrowed Scope:** Narrowing the scope allows attack-specific feature engineering, stronger testing, clearer validation, and a reliable live demonstration.

04 The Three Target Attacks

Clear Visual Explanations of the Network Threat Behaviors Detected by KairosAI



Port Scanning

Reconnaissance

Detect abnormal connection attempts across multiple ports. Attackers systematically probe target systems to identify active services and entry points.

📁 Attacker → Target

- 🔴 Port 21 (SYN) → Closed
- 🔴 Port 22 (SYN) → Closed
- 🟢 Port 80 (SYN) → Open



SYN Flood

Denial of Service

Detect abnormal TCP SYN rates and incomplete connection behavior. Overwhelms systems by sending connection requests and ignoring handshakes.

📁 Web Server Buffer

- 🟡 SYN Recv. (Half-Open)
- 🟡 SYN Recv. (Half-Open)
- 🔴 Queue Full (No Access)



ARP Spoofing

Network Manipulation

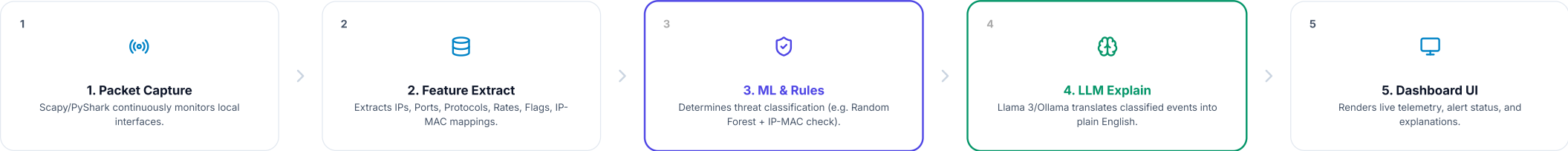
Detect suspicious changes or conflicts in IP-MAC mappings. Associating the attacker's MAC with a local gateway IP intercepts traffic.

📁 Subnet ARP Table

- 🟢 192.168.1.1 → MAC_A
- 🔴 192.168.1.1 → MAC_B (Fake!)
- 🟡 Traffic Intercepted

05 Detection & Explanation Methodology

A Rigorous Pipeline Separating Algorithmic Detection from LLM Explanation



Algorithmic Detection (ML & Rules)

Responsibilities: Determining if an attack is happening based on traffic variables.

- **Port Scan:** Flags connection counts to consecutive destination ports (Random Forest).
- **SYN Flood:** Checks half-open connections and TCP SYN packets/second thresholds.
- **ARP Spoofing:** Stateful rule checks verifying active IP-MAC mapping mismatches.

AI Explanation Layer (Llama 3)

Responsibilities: Translating logs and features into understandable text and recommended actions.

- **No Decision-Making:** The LLM does **not** detect the threat; it only explains it.
- **Plain English Translation:** Converts hex logs, IP strings, and ports to human context.
- **Actionable Intelligence:** Suggests responses (e.g. "Isolate host 192.168.1.105").

06

System Architecture

Unified Technical Data Stream and Component Stacks of the Prototype

 Data Flow Pipeline

- 1

Packet Capture & Feature Extraction
Scapy/PyShark processes raw frames, extracting metadata and tracking state in memory.
- 2

ML & Rule-Based Detection
Feature arrays are run through Scikit-Learn models and static verification rules.
- 3

Alerting & Local AI Explanation
If a threat is classified, a local Ollama Llama 3 context is queried to explain the incident.
- 4

Dashboard UI & PDF Reports
Real-time React interface renders the alert. Users can download a detailed incident report PDF.

 Tech Stack Layers

 **COLLECTION & API**
Scapy • PyShark • Python 3.11 • FastAPI
Asynchronous, high-performance sniffing engine capturing traffic metadata locally.

 **PROCESSING & AI/ML**
Scikit-learn • Random Forest • Isolation Forest • Llama 3 • Ollama
Local feature calculation and models paired with local language intelligence.

 **USER INTERFACE & DB**
React • Chart.js • React Flow • SQLite
Real-time telemetry streaming dashboard, visual flow mapping, and localized database storage.

07 Live Prototype / Demo

Interactive Threat Scenarios: Attack Trigger → Detection → Alert → AI Explanation

TARGET ATTACK

METHOD

RISK

1

Port Scanning

Reconnaissance probing

ML + Rules

MED

2

SYN Flood

Volumetric DDoS flood

ML Model

HIGH

3

ARP Spoofing

Subnet interception

State Rules

HIGH

🔍 Interactive Simulation: Click the attack tabs on the left to see the detection telemetry and local LLM explanation generated for each scan scenario.

http://localhost:8000/alerts/port-scan

> DETECTOR CONSOLE

RECONNAISSANCE DETECTED

Trigger: Active Scanning

🚨

PORT SCAN DETECTED

Source: 192.168.1.105 | Target: 192.168.1.1 (Gateway)

Feature Telemetry:

• Scanned Ports Count: 84 ports in 2.4s

• TCP Connection Flags: SYN only

• Detection Confidence (Random Forest): 94.0%

🧠 Llama 3 AI Explanation: "Host 192.168.1.105 contacted 84 consecutive ports in 2.4 seconds, which is characteristic of network reconnaissance (port scanning). The system suggests isolating the workstation to prevent further asset discovery."

08

Testing & Validation

Rigorous Testing Metrics Measured in Controlled Environments during Round 2 Testing

Attack Vector	Detection Method	Accuracy	Precision	Recall	F1 Score
Port Scanning	Random Forest (ML) + Target Probing Check	[Accuracy]	[Precision]	[Recall]	[F1 Score]
SYN Flood DDoS	SYN Rate Thresholding + TCP State Check	[Accuracy]	[Precision]	[Recall]	[F1 Score]
ARP Spoofing	Subnet IP-MAC Address Cache Rules	[Accuracy]	[Precision]	[Recall]	[F1 Score]

 **False Positive Rate**
Metric: [False Positive Rate]
Measures normal campus activities incorrectly flagged as threats.

 **Detection Inference Latency**
Metric: [Detection Latency]
Measures elapsed time between packet capture and engine classification.

 **Validation Note:** All values above are placeholders and will be updated with measured Round 2 experimental results.

09 Limitations & Scalability

A Realistic Appraisal of Prototype Boundaries and the Roadmap for Future Expansion

Current Scope & Limitations

- **Attack Domain Limits:** Currently limited to detecting Port Scanning, SYN Flood, and ARP Spoofing.
- **Environment Scope:** Verified in controlled, local virtual subnet environments rather than enterprise scale.
- **Response Scope:** Focused on detection, real-time alerting, and LLM explanation. Automated blocking or containment is currently out of scope.
- **ML Model Dependencies:** Anomaly detection thresholds depend heavily on clean local network baseline training datasets.
- **Encrypted Payloads:** Operates strictly on TCP/IP packet header metadata and flow statistics; deep payload inspections are limited by encryption.

Future Scaling Roadmap

- **Attack Class Expansion:** Integrate support for additional classes such as DNS tunneling, credentials brute forcing, and internal spoofing.
- **Distributed Sensor Architecture:** Deploy lightweight physical sensor nodes across multiple campus subnets feeding a central collector.
- **SIEM & Operations Integration:** Forward telemetry into standardized platforms (e.g. syslog, Splunk, Elastic) for unified alerting.
- **Threat Intelligence Feeds:** Pull live reputation updates and malicious IP lists to block known bad actors.
- **Automated Incident Response:** Implement programmatic host isolation and automated firewall rules based on approved playbooks.

10 From Detection to Defense

Refining Intelligent Anomaly Detection & Plain-English Security Explanations



1. DETECT

Focused real-time detection of Port Scan, SYN Flood, and ARP Spoofing using network features and ML-based behavioral checks.



2. EXPLAIN

A local Llama 3 engine translates technical TCP/IP features and logs into plain-English security intelligence and response actions.



3. SCALE

Future roadmap to support additional attack classes, distributed sensors, external threat intelligence feeds, and SIEM integration.



KAIROSAI CONCLUSION

KairosAI turns complex network activity into understandable security intelligence.

DETECT. EXPLAIN. DEFEND.



Engineering Team

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